Uniform second moments of finite-time memory fields
A quantitative bridge from Gaussian chaos to energy functions

Alan Day Chengheng*

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Abstract

We establish uniform annealed second moments for the energy-pinched finite-time memory field in the common-coupling complex double-scaled model. The charged-propagator approximation is first extended to fixed time derivatives and energy insertions. An even Gaussian Dyson expansion then isolates one potentially growing neutral term. A finite-parameter Jacobi recurrence compares coupling-Gaussian Wick powers with deterministic energy polynomials at a quantitative rate. Its entire expansion applies also to the unbounded finite-size Gaussian Hamiltonian, and removes that remaining growth after exact energy pinching. Together with the preceding conserved-return theorem, the result justifies replacing the projected memory field by its ordinary-propagation term in two-point functions. The value and convergence of the limiting fluctuation kernel remain to be calculated.

1 Model and the finite-time problem

Use standard CAR, $\hbar = 1$, string time, and the overlapping complex p-body Gaussian model

$$H = \sum_{I,J} J_{IJ} c_I^\dagger c_J, \quad J = J^\dagger, \qquad \mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \tag{1}$$

$$w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda_N = p^2/N. \tag{2}$$

All charge sectors use the same sample J. Take even $N \to \infty$, $\lambda_- \le \lambda_N \le \lambda_+$ with $\lambda_- > 0$, fixed $J_s > 0$, and compact real memory times $|s| \le S$. The state is the full $\beta = \mu = 0$ normalized annealed trace,

$$\|Y\|_2^2 = \sum_m b_m \mathbb{E}\, \mathrm{tr}_m\, Y_m^\dagger Y_m, \qquad b_m = 2^{-N} \binom{N}{m}, \quad \mathrm{tr}_m = \mathrm{Tr} \,/ \binom{N}{m}. \tag{3}$$

At charge m put $h = H_m$, $h_- = H_{m-1}$, and

$$T(X) = \frac{1}{m} \sum_i c_i^\dagger X c_i, \quad F_m(s) = T(e^{ish_-}), \quad \alpha = (1 - p/m)_+. \tag{4}$$

The variance and a uniform bound are

$$C = C_m = w_N \binom{m}{p} \binom{N - m + p}{p}, \qquad M = \sup_m C_m \le \frac{J_s^2}{\lambda_N} e^{2p/N}. \tag{5}$$

The covariance map $\Phi(X) = \mathbb{E}_u[uXu]$ always uses a new copy u in this charge, with its input held fixed.

*AI research assistant (OpenAI).

2 Memory fields and the result

For $m > 0$, use the exact charge-removal dilation

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \qquad K = 1_N \otimes H_{m-1}, \qquad Q = 1 - VV^\dagger, \tag{6}$$

$$a = V^\dagger K V = \alpha h, \qquad R = QKV, \qquad D = QKQ, \tag{7}$$

$$M_m(s) = R^\dagger e^{isD} R, \qquad \Lambda_m(s) = R^\dagger e^{isK} R. \tag{8}$$

For $m < p$ the force vanishes, so clipping α changes nothing. Energy pinching retains complete degenerate blocks:

$$\mathsf{P}_h Y = \sum_e P_e Y P_e. \tag{9}$$

Define the actual and ordinary-propagation fields

$$\mathsf{F}_m(s) = (1 - \mathsf{P}_h) m e^{-ihs/2} M_m(s) e^{-ihs/2}, \tag{10}$$

$$X_{0,m}(s) = (1 - \mathsf{P}_h) m e^{-ihs/2} \Lambda_m(s) e^{-ihs/2}. \tag{11}$$

The earlier return calculation [7] establishes

$$d_N(S) := \sup_{|s| \le S} \|\mathsf{F}(s) - X_0(s)\|_2 \longrightarrow 0. \tag{12}$$

It did not by itself bound either second moment. The new conclusion is

$$\boxed{\sup_N \sup_{|s| \le S} \|X_0(s)\|_2 < \infty, \qquad \limsup_{N \to \infty} \sup_{|s| \le S} \|\mathsf{F}(s)\|_2 < \infty.} \tag{13}$$

The first supremum is along the specified double-scaling sequence; finitely many initial sizes can be absorbed into its constant. The cited return theorem assumes a convergent λ_N. For a sequence in the stated positive compact interval, every subsequence has a further subsequence with $\lambda_N \to \lambda > 0$; applying that theorem proves (12) on the full sequence.

For $Y = \mathsf{F}$ or X_0, define

$$\mathcal{C}_Y(s_1, s_2; t) = \mathbb{E}\tau\big[Y(s_1)^\dagger e^{itH} Y(s_2) e^{-itH}\big]. \tag{14}$$

Equations (12) and (13) imply

$$\sup_{|s_1|, |s_2| \le S,\ t \in \mathbb{R}} |\mathcal{C}_{\mathsf{F}}(s_1, s_2; t) - \mathcal{C}_{X_0}(s_1, s_2; t)| \to 0. \tag{15}$$

This is an equivalence of the two finite-N correlators, not an evaluation or a proof of convergence of their common large-N limit.

The proof has three parts: weighted transfer to an explicit Gaussian OU force; a finite Dyson expansion retaining its common inserted copy; and a quantitative, exactly conserved energy reference for the neutral covariance sandwich. The last part does not assume a finite-size Hamiltonian has compact spectrum.

3 Time derivatives and energy insertions

On Gaussian matrix-valued L^2 let $\|\psi\|_r = \|(\mathcal{N}+1)^{r/2}\psi\|_2$, $r \geq 0$. This is a Gaussian Sobolev norm, not a Schatten-r norm. For $H = \sum_\beta g_\beta B_\beta$, $\sum B_\beta^2 = CI$, creation and annihilation give, for either left or right multiplication,

$$\|H_{L/R}\psi\|_r \leq A_r\sqrt{C}\|\psi\|_{r+1}, \qquad A_r = 1 + 2^{r/2}. \tag{16}$$

The number-weight energy estimate gives

$$\|e^{itH_L}\psi\|_r \leq e^{q_r\sqrt{C}|t|}\|\psi\|_r, \quad q_r = r2^{\max(r-1,0)}, \quad q_0 = 0. \tag{17}$$

For example, its adjacent-block estimate is $\sqrt{k+1}|(k+2)^r - (k+1)^r|/[(k+1)^{r/2}(k+2)^{r/2}] \leq q_r$.

Recall $T = \mathcal{U}S_- = S_+\mathcal{U}$ from [6], and put

$$n = N - m + 1, \quad B_m = \lambda_N + \lambda_N^2 + \lambda_N(m-n)^2/N, \quad G = \mathcal{U}H_{-,L} - H_{+,L}\mathcal{U}. \tag{18}$$

In central sectors $N/4 \leq m \leq 3N/4$, take $N \geq 16$, $p \leq N/16$. The low-chaos defect is bounded by $10^6\sqrt{CB_m}(k+1)^{3/2}/N$. In the complementary block, $(k+1)p > N/8$ implies $N \leq 64\lambda_N(k+1)^2$. Thus the coarse Gaussian-field bound absorbs the entire high block, and, with harmless loose constants,

$$\|G\psi\|_r \leq 10^7 A_r\frac{\sqrt{MB_m}}{N}\|\psi\|_{r+5}, \tag{19}$$

$$\|[\mathfrak{D}, H_L]\psi\|_r \leq 10^4 A_r\sqrt{M(\lambda_N + \lambda_N^2)}\|\psi\|_{r+3}, \qquad \mathfrak{D} = p\mathcal{N} - \mathcal{R}. \tag{20}$$

The second line is the all-degree rank estimate [5] with number weights. For $\delta_j(s) = \mathcal{U}e^{ish_-}h_-^j - e^{ish}h^j$, the initial difference is exactly $\sum_{a<j} H_{+,L}^{j-1-a}GH_{-,L}^a I$. Weighted Duhamel using (17) therefore controls δ_j by $K_{r,j,S,M}\sqrt{B_m}/N$. Similarly, starting from $\mathfrak{D}H^jI = \sum_{a<j} H_L^a[\mathfrak{D}, H_L]H_L^{j-1-a}I$ gives $\sup_{|s|\leq S}\|\mathfrak{D}h^je^{ish}\|_r \leq K\sqrt{\lambda_N + \lambda_N^2}$.

Apply the preceding exact rank-filter inequality to h^je^{ish}:

$$\|(S_+ - \alpha^{\mathcal{N}})\phi\|_r \leq \frac{4}{N}\|\mathfrak{D}\phi\|_r + \frac{8p|m-n|}{N^2}\|\phi\|_{r+2} + \frac{528p^2}{N^2}\|\phi\|_{r+4}. \tag{21}$$

For all fixed r, j, a, b, the actual derivative estimate is consequently

$$\boxed{\sup_{|s|\leq S}\|h^a\partial_s^j[F_m(s) - \mathcal{O}_\alpha(e^{ish})]h^b\|_r \leq K_{r,j,a,b,S,M}\frac{\sqrt{B_m}}{N}.} \tag{22}$$

Here $\mathcal{O}_\alpha = \alpha^{\mathcal{N}}$. Finite Hermite polynomials provide a common core; finite-chaos energy estimates, closure and Duhamel give the stated domains. No derivative of an unweighted error estimate was taken.

4 An even Gaussian Dyson expansion of the force

The exact derivative identity and (22) give

$$\Lambda = -F_m'' + i\alpha\{h, F_m'\} + \alpha^2 h F_m h, \tag{23}$$

$$\|\Lambda - \Lambda_{\mathrm{OU}}\|_2 \le K_S \sqrt{B_m}/N, \tag{24}$$

$$\Lambda_{\mathrm{OU}}(s) = \epsilon \mathbb{E}_u[u e^{is(\alpha h + \sqrt{\epsilon} u)} u], \qquad \epsilon = 1 - \alpha^2. \tag{25}$$

The same u occurs inside and outside the exponential. Fix h and write $U_t = e^{i\alpha h t}$. The function $f(t) = \mathbb{E}_u[u e^{is(\alpha h + tu)} u]$ is even. Taylor expansion in $t = \sqrt{\epsilon}$, keeping αh exact, yields

$$\Lambda_{\mathrm{OU}}(s) = \epsilon \Phi(U_s) - \epsilon^2 \mathcal{I}_s(h) + \mathcal{E}_s, \tag{26}$$

where, for either sign of s,

$$\mathcal{I}_s(h) = s^2 \int_{0 \le y \le x \le 1} \mathbb{E}_u[u U_{s(1-x)} u U_{s(x-y)} u U_{sy} u]\, dx\, dy. \tag{27}$$

All three Wick pairings are retained. For unitary A, B, D,

$$\mathbb{E}_u[uAuBuDu] = \Phi(A)B\Phi(D) + \Phi(A\Phi(B)D) + \sum_{\beta,\gamma} B_\beta A B_\gamma B B_\beta D B_\gamma. \tag{28}$$

Each term has operator norm at most C^2. For the last term use row–column Cauchy–Schwarz and $\sum_\beta \Phi(BB_\beta D)^\dagger \Phi(BB_\beta D) \le C^3 I$. It follows that

$$\|\mathcal{I}_s(h)\|_\infty \le \tfrac{3}{2} C^2 s^2. \tag{29}$$

For completeness, Gaussian integration by parts and weighted Young's inequality in a spectral basis of u give

$$\mathbb{E}\,\mathrm{tr}\, u^{2q} \le (2q-1) C \mathbb{E}\,\mathrm{tr}\, u^{2q-2} \le (2q-1)!! C^q. \tag{30}$$

This is the scalar-variance case of standard Gaussian matrix moment control; compare [1]. Every fourth Duhamel derivative of f contains six factors u and otherwise real-time unitaries. Normalized Schatten-12 Hölder, followed by Jensen, therefore gives the dimension-free, conditional-on-h bound

$$\boxed{\|\mathcal{E}_s\|_2 \le \frac{\epsilon^3 |s|^4}{24} \sqrt{11!!}\, C^3.} \tag{31}$$

In central sectors $m\epsilon \le 2p$, $m\epsilon^2 \le 16\lambda_N$, and $m\epsilon^3 \le 128\lambda_N^2/p$. Thus the second term of (26) is uniformly bounded at collective scale and the remainder vanishes. Only $p(1 - \mathsf{P}_h)\Phi(U_s)$ can still grow.

5 A finite-parameter annihilation recurrence

We now control that neutral term. All statements in this section are in a central charge sector. Let

$$Z_j = \mathsf{W}_j(H^j), \qquad \mu(x) = \prod_{\ell=0}^{p-1} \left(1 - \frac{x}{m-\ell}\right)_+ \left(1 - \frac{x}{N-m+p-\ell}\right)_+, \quad a_j = \sum_{v=0}^{j-1} \mu(vp). \tag{32}$$

W_j is the coupling-Gaussian chaos projection, not spectral orthogonalization. The deterministic covariance satisfies $\Phi(P_r X) = C\mu(r)P_r X$, and μ is globally $12p/N$-Lipschitz. With Gaussian creation $\mathcal{C}$ and annihilation $\mathcal{A} = \mathcal{C}^\dagger$, $Z_j = \mathcal{C}^j I$. Differentiating the homogeneous matrix polynomial before projecting gives the exact identity

$$\mathcal{A}Z_j = \mathsf{W}_{j-1}\left[\sum_{v=0}^{j-1} \Phi(H^v)H^{j-1-v}\right] = Ca_j Z_{j-1} + r_j. \tag{33}$$

No Wick contractions are omitted; all derivative positions are the v-sum in (33).

Let π_v be the positive rank law of a product of v independent Hamiltonians and $D_v = vp - r$. The earlier all-degree estimate is

$$\mathbb{E}_{\pi_v} D_v^2 \le 2 \cdot 10^6(\lambda_N v^2 + \lambda_N^2 v^4). \tag{34}$$

For a fixed derivative position, apply the Lipschitz estimate to the first v independent tensor factors. The remaining factors are eliminated in the norm using $\mathbb{E}H_i^2 = CI$. Gaussian symmetrization is a contraction. Hence

$$\boxed{\|r_j\|_2 \le \frac{12p}{N}\sqrt{(j-1)!\,C^{j+1}} \sum_{v<j} \sqrt{\mathbb{E}_{\pi_v} D_v^2}.} \tag{35}$$

There is only one $(j-1)!$. This estimate is uniform in degree, including vp beyond the physical rank range.

Retaining finite-N coefficients is useful. Bernoulli's inequality in each positive factor gives

$$\mu(jp) \le \mu(p)^j, \qquad \mu(p) \le \exp\left[-\frac{4\lambda_N}{1+p/N}\right]. \tag{36}$$

The second estimate uses the reciprocal sum of the two denominators, which is at least $4/(N+p)$. Thus for all $j \ge 1$,

$$1 \le a_j \le A, \qquad A = (1 - e^{-3\lambda_-})^{-1}, \tag{37}$$

under $p \le N/16$. We do not replace $\mu(jp)$ by its leading q^j limit in the quantitative argument.

6 Auxiliary Jacobi polynomials and the chaos error

Define monic scalar polynomials

$$Q_0(x) = 1, \quad Q_1(x) = x, \qquad Q_{j+1}(x) = xQ_j(x) - Ca_jQ_{j-1}(x). \tag{38}$$

They belong to an auxiliary positive Jacobi problem [2], whose off-diagonal entries are $\sqrt{Ca_j}$ and whose diagonal is zero. It has norm at most $2\sqrt{CA}$. This auxiliary measure is not asserted to be the actual finite-N annealed density of states.

Put $E_j = Z_j - Q_j(H)$. The leading coupling degree cancels, and parity leaves degree at most $j-2$. Equations (33) and (38) give

$$E_{j+1} = HE_j - Ca_jE_{j-1} - r_j, \qquad \|HE_j\|_2 \le 2\sqrt{C(j-1)}\|E_j\|_2. \tag{39}$$

Write $\rho = p/N$ and

$$e_j = \frac{\|E_j\|_2}{\sqrt{j!}\,C^{j/2}}, \quad L = 12\sqrt{2\cdot 10^6}(\sqrt{\lambda_+} + \lambda_+), \quad K_* = 4 + A. \tag{40}$$

The rank moment bound and (35) imply

$$e_{j+1} \le 2e_j + \frac{A}{\sqrt{j(j+1)}}e_{j-1} + \rho L j^2. \tag{41}$$

Since $e_0 = e_1 = e_2 = 0$, a direct induction using $j^2 \le 3^j$ and $K_* \ge 5$ yields

$$\boxed{\|Z_j - Q_j(H)\|_2 \le \rho L K_*^j \sqrt{j!}\,C^{j/2}.} \tag{42}$$

Also $\|Z_j\|_2 \le \sqrt{j!}\,C^{j/2}$ by independent-tensor symmetrization. No uniform lower bound on C is required; the case $C = 0$ is simply $H = 0$.

Let J_{aux} be the bounded Jacobi operator and define

$$h_j = C^j \prod_{\ell=1}^{j} a_\ell, \qquad c_j(s) = \frac{\langle e_j, e^{isJ_{\mathrm{aux}}}e_0\rangle}{\sqrt{h_j}}. \tag{43}$$

Bandedness implies that the numerator's Taylor series starts at degree j. Since $h_j \ge C^j$, for $|s| \le S$,

$$|c_j(s)| \le e^{2\sqrt{CA}S}\frac{(2\sqrt{A}S)^j}{j!}. \tag{44}$$

This factorial decay beats the $\sqrt{j!}$ in (42). It is the required control of the order of limits: the full degree sum is controlled before the large-N limit.

7 Entire expansion and an exactly conserved reference

The Jacobi expansion is valid outside its auxiliary spectral interval. The recurrence implies $|Q_j(z)| \le (|z|+\sqrt{CA})^j$. Together with (44), this makes $f_s(z) = \sum_j c_j(s)Q_j(z)$ normally convergent on every complex compact set. The Jacobi equation gives

$$c_j'(s) = i[c_{j-1}(s) + Ca_{j+1}c_{j+1}(s)], \qquad c_{-1} = 0. \tag{45}$$

Termwise differentiation and (38) yield $\partial_s f_s(z) = izf_s(z)$, $f_0(z) = 1$. Therefore

$$e^{isz} = \sum_{j=0}^{\infty} c_j(s)Q_j(z), \qquad z \in \mathbb{C}. \tag{46}$$

It applies to every finite Gaussian sample's spectrum. Separately, (42) implies absolute convergence in the actual annealed matrix L^2. Thus no spectral cutoff has been imposed on H.

One explicit uniform summation constant is

$$R_S = \sqrt{2}L\exp\left(2\sqrt{AM}S + 4K_*^2AMS^2\right), \tag{47}$$

using $\sum_j x^j/\sqrt{j!} \le \sqrt{2}e^{x^2}$. Consequently

$$\left\| e^{isH} - \sum_j c_j(s)Z_j \right\|_2 \le \rho R_S. \tag{48}$$

For any deterministic sequence $|\theta_j| \le 1$, define

$$g_{\theta,s}(H) = \sum_j \theta_j c_j(s)Q_j(H). \tag{49}$$

This is an actual samplewise energy function, not an assertion that a chaos multiplier commutes with H. Contraction of the multiplier and a second use of (42) give

$$\boxed{\|\theta(\mathcal{N})e^{isH} - g_{\theta,s}(H)\|_2 \le 2\rho R_S.} \tag{50}$$

θ may depend on N, m, s, but not on the original random J.

Now choose $\theta_j = \mu(jp)$. The earlier rank–chaos filter estimate is

$$\|(\Phi/C - \theta(\mathcal{N}))e^{isH}\|_2 \le 12\rho\|\mathfrak{D}e^{isH}\|_2. \tag{51}$$

Let D_S denote its already established uniform bound on $\|\mathfrak{D}e^{isH}\|_2$ [5]. Since $g_{\theta,s}(H)$ commutes with H and pinching is an orthogonal projection, we obtain

$$\boxed{\sup_{|s|\le S} p\|(1-\mathsf{P}_h)\Phi(e^{isH})\|_2 \le C\lambda_N(12D_S + 2R_S).} \tag{52}$$

This is the needed quantitative energy subtraction. The auxiliary Jacobi and coupling-Gaussian constructions were never identified exactly at finite N; their difference was explicitly bounded.

8 Assembly of the true memory bound

Within a central sector, combine (24), (26), (31) and (52). The phases $e^{-ihs/2}$ are unitary and commute with the pinching. As $m\epsilon/p \le 2$, one obtains a bound of the form

$$\sup_{|s|\le S} \|X_{0,m}(s)\|_2 \le K_S\sqrt{B_m} + 2M\lambda_N(12D_S + 2R_S) + 24\lambda_N M^2 S^2 + \frac{128\lambda_N^2}{p}\frac{S^4\sqrt{11!!}M^3}{24}. \tag{53}$$

The first term is the collective transfer defect, and the third is the four-u correction. Both can be order one; neither has been discarded. The last term tends to zero.

The full-charge average satisfies

$$\sum_m b_m(m-n)^2 = N+1, \qquad \sum_{m\ \text{central}} b_m B_m \le \lambda_N + \lambda_N^2 + \lambda_N(1+1/N). \tag{54}$$

Squaring (53) and summing therefore stays bounded. For the exterior sectors, T commutes with the Gaussian number operator and is an L^2 contraction. The all-sector variance bound and (17) give uniform coarse bounds on the fixed derivatives and energy insertions in Λ; multiplying by $m \le N$ costs at most N^2 in the squared norm. The exterior binomial weight is at most $2e^{-N/8}$, so this contribution vanishes. This proves the first assertion of (13); the second follows from (12).

Finally Hilbert–Schmidt Cauchy–Schwarz gives, uniformly in real t,

$$|\mathcal{C}_{\mathsf{F}} - \mathcal{C}_{X_0}| \le \|\mathsf{F}(s_1) - X_0(s_1)\|_2\|\mathsf{F}(s_2)\|_2 + \|X_0(s_1)\|_2\|\mathsf{F}(s_2) - X_0(s_2)\|_2. \tag{55}$$

The right side is bounded by a constant times $d_N(S)$, proving (15).

Next calculation. Uniform second moments now hold for the actual finite-memory field. The remaining task is the joint limit of the bounded components: the charged-transfer defect, the neutral covariance-filter residual and the four-u correction, with the exact common disorder and energy pinching retained. Their cross terms can contribute at order one. One must compute these terms to determine the finite-time fluctuation kernel and compare it to the Susskind meson dictionary. The present bound alone does not supply that kernel or a finite-G bulk interaction law.

9 Verification and sources

Three fresh scientific outputs accompany the proof. Complete Gaussian-Hermite coefficients in three CAR systems verify twelve annihilation recurrences through degrees 5, 4 and 3, including the lower-chaos terms of $Q_j(H)$. An independent covariance-word construction checks (33); positive rank tensors check the single-factorial remainder normalization. A full four-coordinate Hermitian 2×2 Gaussian fixture checks three fourth-moment Wick words and nine noncommuting finite-time Dyson remainders. Two Gaussian and time quadrature orders were compared. These order comparisons are not certified quadrature enclosures.

The auxiliary Jacobi series was also evaluated at $x = \pm 6$, outside its spectral bound, for three finite parameter sets at 80 and 120 digits. Six checks and 105 exact geometric-weight inequalities are saved. These confirm finite formulas and cancellation conventions; the uniform bounds and entire continuation are established by the arguments above, not by finite fits. Inputs, environments, hashes, build logs and rendered pages are saved locally. Old scientific outputs are unchanged.

Literature boundary. Gaussian matrix moments, Jacobi theory and q-Gaussian conditional expectation are established tools [1, 2, 3]. The spectral Wick polynomials in [4] are defined by annealed spectral orthogonality. They are not identical at finite N to $\mathsf{W}_j(H^j)$; (42) supplies the missing comparison for this argument, using an auxiliary finite-parameter recurrence. We do not infer priority from a literature search. The new local result concerns the specified microscopic field and its second moments, with the prior-stage estimates listed below as dependencies.

References

[1] J. A. Tropp, *Second-Order Matrix Concentration Inequalities*, Prop. 2.3 and Sec. 7, author-hosted preprint (2016).

[2] NIST Digital Library of Mathematical Functions, Sec. 18.2, general orthogonal polynomials.

[3] M. Bożejko, B. Kümmerer and R. Speicher, *q-Gaussian processes: non-commutative and classical aspects*, arXiv:funct-an/9604010, Prop. 2.9 and Theorem 2.11.

[4] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 3.1.

[5] A. Day and Chengheng, *Uniform rank bounds for finite-time propagation*, local research note (7 November 2026).

[6] A. Day and Chengheng, *Adjacent-charge propagation as Gaussian smoothing*, local research note (7 November 2026).

[7] A. Day and Chengheng, *The conserved background of the first memory return*, local research note (6 November 2026).